



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/12

Paper 1 Further Pure Mathematics 1

May/June 2025

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

- 1 (a) Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n (2-3r)(5-3r) = an^3 + bn^2 + cn,$$

where a , b and c are integers to be determined. [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b) Use the method of differences to find $\sum_{r=1}^n \frac{1}{(2-3r)(5-3r)}$ in terms of n . [4]

This image shows a full page of white paper designed for handwriting practice. It features approximately 20 evenly spaced horizontal dotted lines running across the entire width of the page. There are no margins, text, or other markings present.

- (c) Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{(2-3r)(5-3r)}$. [1]

2 The cubic equation $x^3 + 2x + 1 = 0$ has roots α, β, γ .

(a) Find a cubic equation whose roots are $\alpha^3 - 1, \beta^3 - 1, \gamma^3 - 1$. [3]

This image shows a full page of a document template. It consists of approximately 28 horizontal dotted lines spaced evenly down the page, providing a guide for handwriting practice or structured note-taking. The lines are light gray and extend across the entire width of the page. There is no text, imagery, or other markings present.



(b) Find the value of $(\alpha^3 - 1)^2 + (\beta^3 - 1)^2 + (\gamma^3 - 1)^2$. [2]

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(c) Find the value of $(\alpha^3 - 1)^3 + (\beta^3 - 1)^3 + (\gamma^3 - 1)^3$. [2]

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3 The sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = 5$ and $u_{n+1} = 6u_n + 5$ for $n \geq 1$.

(a) Prove by induction that $u_n = 6^n - 1$ for all positive integers n . [5]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.

(b) Deduce that u_{2^n} is divisible by u_n for $n \geq 1$. [2]

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4 The matrix $\mathbf{M}$ is given by $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, where $0 < \theta < 2\pi$.

(a) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations in the x - y plane.

State the type of each transformation, and make clear the order in which they are applied. [2]

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(b) Find the value of θ for which the transformation represented by $\mathbf{M}$ has a line of invariant points. [7]

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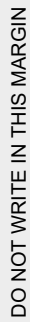
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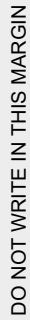
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- $$\theta \cos \theta + \left(\frac{1}{8}\theta + 1\right) \sin \theta = 0$$

and verify that this equation has a root between 5 and 5.05. [5]



6 The points A, B, C have position vectors

$$\mathbf{i} - 2\mathbf{k}, \quad \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}, \quad 2\mathbf{i} - \mathbf{j} - \mathbf{k},$$

respectively.

(a) Find the equation of the plane ABC , giving your answer in the form $ax + by + cz = d$. [5]

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A point D has position vector $\mathbf{i} + t\mathbf{k}$, where $t \neq -2$.

(b) Find the acute angle between the planes ABC and ABD . [4]

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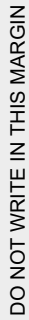
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[2]

[illegible]

[4]

[illegible]



(c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

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(d) Sketch the curve with equation $y = \left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right|$.

[1]



- (e) Find in exact form the set of values of x for which $\left| \frac{2x^2 - 5x}{2x^2 - 7x - 4} \right| < \frac{1}{9}$. [5]

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Additional page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

Lined area for writing answers, consisting of multiple horizontal dotted lines.





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